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Dominic Duxbury, PhD

A Data Consultant and ex-AI researcher with experience working on real-time decision support systems and state of the art ML techniques. I have delivered software for pathogen characterisation, algorithms for drones, public sector data migrations and petabyte-scale data audits.

Experience

Data Architect | Marvel Consulting (Contract)

February 2025 - April 2025 | Department of Health and Social Care (DHSC)

For this discovery project I worked within a team at Marvel Consulting, analysing ways forward for the **Adult Social Care Workforce Data Set (ASC-WDS)** and workforce data as a whole at the DHSC. I worked alongside a technical architect, splitting responsibilities between data and systems processes. We carried out technical discovery interviews together and drew up documentation detailing the current system, moving on to create a vision of a future system, running on the DHSC in-house cloud within **Azure**.

Senior Data Consultant | Hippo (Contract)

December 2024 - August 2025 | Pathogena EIT

In this role I gained experience leading a team in a small biotech start-up environment working in **Oracle Cloud**. A small team from Hippo has been working to productionise cutting-edge science from a group of Oxford scientists. This worked through a modern **django** app using a **ReactJS frontend** to pass DNA data files through to a **Kubernetes** cluster of data processors. I contributed towards the release of a pipeline for the automatic characterisation of new variants of SARS-CoV-2, my role included scoping pieces of work, ticketing and various other facets of managing a small team of fullstack engineers.

April 2024 - December 2024 | Flutter Int.

A data retention workstream was spun up at Flutter Int. as a small but important part of their large scale Cloud Transformation project. I completed parts of the discovery for this workstream, with the objective of reviewing and analysing databases within the Flutter cloud estate which uses a patchwork of cloud technologies. The databases were judged with regard to Flutter's data retention policy, to ensure GDPR is followed and all risks are highlighted. The findings were then reported to senior stakeholders through a series of discovery documentation. Over **7 TB** of data was highlighted to be deleted of the **15+ TB** surveyed. My role included developing strategies to extract and analyse data plus documentation of acquired data and decisions made.

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February 2024 - April 2024 | DHSC

Subcontracted at the Department of Health and Social Care alongside a Hippo team as part of a departmental data migration into **Azure**. We had 8 weeks to discover and document 5 systems, then present a migration plan for the products. My role included interviewing product owners, lending data engineer knowledge to the creation of new architectures and applying options analysis techniques from my PhD to highlight trade-offs between different approaches. Finally, I was part of the team delivering the final pitch directly to stakeholders.

Senior Data Engineer | Netacea

February 2022 - December 2023

I have led the development of a new platform for delivering real-time models at Netacea. This platform has leveraged pyFlink to create a new modular strategy for delivering data products that can scale up to **billions** of requests per day. The platform fits onto **AWS** as part of a data mesh strategy to enable self-serve data and standardised consumption of high velocity data throughout the business. Leadership experience included creating tickets, planning sprints and leading meetings to prioritise work for a team of six. Developer experience included: applying **python** and **docker** to create a streamlined developer experience with the highlight being a containerised **Kafka/pyFlink** environment for integration tests.

Data Scientist | Laterooms.com

June 2015 - December 2017

I joined Laterooms as a 3-month industrial placement initially, then moved to a permanent role for a total of two years, working as part of a small data science team to identify areas of opportunity, design, and implement solutions, including part-time work whilst I finished my degree. I worked with **Google Cloud Platform** including **BigQuery** for big data analysis.

- Created architecture to allow the comparison of mobile and website behaviour in real-time to help stakeholders understand how best to cater to mobile customers whilst this customer segment grew and the Laterooms app was developed.
- Worked with infrastructure teams to restructure and centralise event streams and databases using **Kafka**, improved deployments and scalability using **Docker**, and utilised microservices architecture as a means to integrate with **Google Cloud Platform**. Allowing us to deliver real value from modern artificial intelligence methodologies.
- Contributed towards the overhaul of the search ranking system. This was a multi-team effort to build a new search system, to allow increased personalisation, new revenue channels, and optimisation of existing KPIs. Leading to **£0.90** more commission per search, a **21%** increase.

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Education

PhD, Dynamic Decision Making | University of Manchester

Sponsored by BAE Systems

October 2017 - October 2021

As part of a PhD, I investigated methodologies for combining stream processing engines with decision support algorithms to enable the production of dynamic decision support systems. The research was proposed by BAE Systems, to enhance decision making in dynamic command and control scenarios. I was focused on high-stakes domains where a human is required to make the final call, equipping these systems with explainable and trustable AI.

- A framework for **dynamic decision making** was created then illustrated using a train journey case study.
- Software supporting the planning of routes for drones was produced. These drones were utilised to improve situational awareness in a harbour.
- A cloud architecture including a **Flink** cluster on **Google Cloud Platform** was used to deliver a user experiment during coronavirus restrictions. This architecture enabled novel decision making algorithms to be executed in real-time.
- Metrics were collected and analysed using **ETL** techniques across this architecture to quantitatively assess the effect of decision support features on trust and its antecedents.
- A headless version of this software was designed to provide comparison between algorithms for **decision support.**

BSC, Computer Science and Mathematics | University of Manchester

October 2013 - June 2017 - **BA (Hons) 1st Class.**

For my mathematics options I focused on statistics and pure mathematics, including group theory and information theory.

- As a final year project I analysed the effect of twitter sentiment on the price of bitcoin. A pipeline was produced, using the twitter API and **NLP** techniques to extract features from tweets.
- These features were used to classify the sentiment of tweets in real-time, and the number of positive/negative tweets were used along with other predictors to forecast small changes in bitcoin price.

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A Levels | Cardinal Newman College, Preston

2011 - 2013

AAA in Maths, Further Maths and Chemistry
B (79%) in AS Level Economics

GCSE | Holy Cross Catholic College, Chorley

2006 - 2011

6 A*, including 3 A* in sciences and an A* in maths
5 A, including English

Publications

Dynamic Decision Making for Situational Awareness using Drones

Expert Systems with Applications - Volume 252, Part B, 15 October 2024, 124057

https://www.sciencedirect.com/science/article/abs/pii/S0957417424009230

In this paper we present five desiderata for dynamic decision support and we compare methods with regards to the consistency of trade-offs between criteria and the stability of results under small changes to criteria values.

Trusted and Auditable Decision Aids over Data Streams

Proceedings of the Workshops of the EDBT/ICDT 2019 Joint Conference

https://ceur-ws.org/Vol-2322/dsi4-3.pdf

This paper investigates approaches for bringing together stream management and decision support systems in a way that is both auditable and trustable.

Software Preferences

Languages: Python, TypeScript, Java, Rust.
Cloud Provider (in order): AWS, GCP, Azure (*Azure Fundamentals Certified*) .
Clouds Development: Docker, Terraform.
Web Development: React, Redux.
Databases: PostgreSQL, MySQL.
Streaming Software: Flink, Arroyo, Kafka.
Machine Learning: Scikit-learn, Matplotlib, TensorFlow, Keras.